

6032A Solar Power Technologies

POLY PMNA - Polytechnic Study Hub

Semester 6 | Program Elective | Credits 4 | Periods 60

Downloadable handbook generated from lessons-6032A.html

Core areas: solar thermal, CSP, PV cells, rooftop arrays, inverters, net metering and grid synchronization.

Course Overview and Roadmap

Objectives: adopt efficient solar technology, extract electrical power or heat using solar technologies, and select/erect solar PV systems.

CO1: Explain technologies to harness solar heat. CO2: Classify CSP plants.

CO3: Explain rooftop solar module array configurations and maintenance.

CO4: Outline solar PV operation and grid synchronization.

Roadmap: M1 solar thermal devices; M2 CSP plants; M3 PV modules and arrays; M4 PV system components, inverters, controllers, net metering and synchronization.

Module Chapters

Module 1: Solar energy advantages, disadvantages, applications, collectors, solar cookers, water heaters, stills, air heaters, dryers, kilns, pumps, greenhouses, space heating and solar absorption cooling.

Module 2: CSP plants by temperature. Low temperature: solar pond and flat plate collector plant. Medium temperature: parabolic collector plant. High temperature: parabolic dish, central receiver, Fresnel reflector and Stirling engine.

Module 3: PV effect, cell construction, crystalline silicon, thin film and emerging PV technologies, V-I characteristics, I_{sc} , V_{oc} , P_{max} , V_{mp} , I_{mp} , fill factor, equivalent circuit, strings and arrays, module specifications and maintenance.

Module 4: PV modules, batteries, inverter, charge controller, mounting, cables, blocking/bypass diode, junction box, buck/boost converters, MPPT/PWM, off-grid, grid-tied and hybrid systems, net metering and grid synchronization.

Formula Bank, Practice and Answers

$P = V \times I$. $P_{max} = V_{mp} \times I_{mp}$. Fill factor = $P_{max} / (V_{oc} \times I_{sc})$. Energy = Power x sunshine hours x derating factor.

Example: $V_{mp} = 36 \text{ V}$ and $I_{mp} = 8 \text{ A}$ gives $P_{max} = 288 \text{ W}$.

Example: $V_{oc} = 44 \text{ V}$, $I_{sc} = 9 \text{ A}$, $V_{mp} = 36 \text{ V}$, $I_{mp} = 8 \text{ A}$. Fill factor = $288 / (44 \times 9) = 0.727$.

Expected questions: explain solar collectors, CSP classification, solar cell V-I curve, array configurations, MPPT, net metering and grid synchronization.

Model paper answer guide: draw diagrams, write formulas, substitute values, state units, mention shading, tilt, direction, temperature and maintenance.